OTHER MAJOR RETURN MEASURES AND THEIR APPLICATIONS

Learning Outcome

- calculate and interpret major return measures and describe their appropriate uses

The statistical measures of return discussed in the previous section are generally applicable across a wide range of assets and time periods. Special assets, however, such as mutual funds, and other considerations, such as taxes or inflation, may require more specific return measures.

Although it is not possible to consider all types of special measures, we will discuss the effect of fees (gross versus net returns), taxes (pre-tax and after-tax returns), inflation (nominal and real returns), and the effect of **leverage**. Many investors use mutual funds or other external entities (i.e., investment vehicles) for investment. In those cases, funds charge management fees and expenses to the investors. Consequently, gross and net-of-fund-expense returns should also be considered. Of course, an investor may be interested in the net-of-expenses after-tax real return, which is in fact what an investor truly receives. We consider these additional return measures in the following sections.

Gross and Net Return

A gross return is the return on assets managed less any trading expenses and commissions. Gross return is intended to reflect the investment skill of the manager. Expenses including management fees, custody fees, and taxes are not included in the gross return because they may be different for different investors. For example, most asset managers provide lower management fee rates to larger accounts. Excluding these expenses in gross returns provides a basis for evaluation and comparison of investment management skill.

Net return is a measure of what the investment vehicle (e.g., mutual fund) has earned for the investor. Net return accounts for (i.e., deducts) all managerial and administrative expenses that reduce an investor's return. Because individual investors are most concerned about the net return (i.e., what they actually receive), small mutual funds with a limited amount of assets

expenses over a larger asset base. As a result, many small mutual funds waive part of the expenses to keep the funds competitive.

Pre-Tax and After-Tax Nominal Return

All return measures discussed up to this point are pre-tax nominal returns—that is, no adjustment has been made for taxes or inflation. In general, all returns are pre-tax nominal returns unless they are otherwise designated.

Many investors are concerned about the possible tax liability associated with their returns because taxes reduce the net return that they receive. Capital gains and income may be taxed differently, depending on the jurisdiction. Capital gains come in two forms: short-term capital gains and long-term capital gains. Long-term capital gains receive preferential tax treatment in a number of countries. Interest income is taxed as ordinary income in most countries. Dividend income may be taxed as ordinary income, may have a lower tax rate, or may be exempt from taxes depending on the country and the type of investor. The after-tax nominal return is computed as the total return minus any allowance for taxes on dividends, interest, and realized gains. Bonds issued at a discount to the par value may be taxed based on accrued gains instead of realized gains.

Because taxes are paid on realized capital gains and income, the investment manager can minimize the tax liability by selecting appropriate securities (e.g., those subject to more favorable taxation, all other investment considerations equal) and reducing trading turnover. Therefore, taxable investors evaluate investment managers based on the after-tax nominal return.

Real Returns

Previously the learning model approximated defined the relationship between the nominal rate and the real rate by the following relationship equation:

$$(1 + nominal\ risk\text{–}\ free\ rate) = (1 + real\ risk\text{–}\ free\ rate)(1 + infl$$

This relationship can be extended to link the relationship between nominal and real returns. Specifically, the nominal return consists of a real risk-free rate of return to compensate for postponed consumption; inflation as loss of purchasing power; and a risk premium for assuming risk. Frequently, the real risk-free return and the risk premium are combined to arrive at the real “risky” rate and is simply referred to as the real return, or:

$$(1 + \text{real return}) = (1 + \text{real risk–free rate})(1 + \text{risk premium}).$$

returns are also useful in comparing returns among countries when returns are expressed in local currencies instead of a constant investor currency and when inflation rates vary between countries (which are usually the case).

Finally, the after-tax real return is what the investor receives as compensation for postponing consumption and assuming risk after paying taxes on investment returns. As a result, the after-tax real return becomes a reliable benchmark for making investment decisions. Although it is a measure of an investor's benchmark return, it is not commonly calculated by asset managers because it is difficult to estimate a general tax component applicable to all investors. For example, the tax component depends on an investor's specific taxation rate (marginal tax rate), how long the investor holds an investment (long-term versus short-term), and the type of account the asset is held in (tax-exempt, tax-deferred, or normal).

EXAMPLE 14

Computation of Special Returns

Let's return to Example 8. After reading this section, Mr. Lohrmann decided that he was not being fair to the fund manager by including the asset management fee and other expenses because the small size of the fund would put it at a competitive disadvantage. He learns that the fund spends a fixed amount of EUR500,000 every year on expenses that are unrelated to the manager's performance.

Mr. Lohrmann has become concerned that both taxes and inflation may reduce his return. Based on the current tax code, he expects to pay 20 percent tax on the return he earns from his investment. Historically, inflation has been around 2 percent and he expects the same rate of inflation to be maintained.

1. Estimate the annual gross return for the first year by adding back the fixed expenses.

Hide Solution

Solution:

The gross return for the first year is higher by 1.67 percent (= EUR500,000/EUR30,000,000) than the 15 percent% investor return reported by the fund. Thus, the gross return for the first year is 16.67 percent (= 15% + 1.67%).

Hide Solution

Solution:

The investor return reported by the mutual fund is the net return of the fund after accounting for all direct and indirect expenses. The net return is also the pre-tax nominal return because it has not been adjusted for taxes or inflation. From Example. 8, the net return for the five-year holding period was calculated as 42.35 percent.

3. What is the after-tax net return for the first year that investors earned from the Rhein Valley Superior Fund? Assume that all gains are realized at the end of the year and the taxes are paid immediately at that time.

Hide Solution

Solution:

The net return earned by investors during the first year was 15 percent. Applying a 20 percent tax rate, the after-tax return that accrues to investors is 12 percent [= 15% − (0.20 × 15%)].

4. What is the after-tax real return that investors would have earned in the fifth year?

Hide Solution

Solution:

The after-tax return earned by investors in the fifth year is 2.4 percent [= 3% − (0.20 × 3%)]. Inflation reduces the return by 2 percent so the after-tax real return earned by investors in the fifth year is 0.39 percent, as shown:

$$\frac{(1+2.40\%)}{(1+2.00\%)} - 1 = \frac{(1+0.0240)}{(1+0.0200)} - 1 = 1.0039 - 1 = 0$$

Note that taxes are paid before adjusting for inflation.

Leveraged Return

In the previous calculations, we have assumed that the investor's position in an asset is equal to the total investment made by an investor using his or her own money. This section differs in that the investor creates a leveraged position.

investor may trade futures contracts in which the money required to take a position may be as little as 10 percent of the notional value of the asset. In this case, the leveraged return, the return on the investor's own money, is 10 times the actual return of the underlying security. Both the gains and losses are amplified by a factor of 10.

Investors can also invest more than their own money by borrowing money to purchase the asset. This approach is easily done in stocks and bonds, and very common when investing in real estate. If half (50 percent) of the money invested is borrowed, then the gross return to the investor is doubled, but the interest to be paid on borrowed money must be deducted to calculate the net return.

Using borrowed capital, debt, the size of the leveraged position increases by the additional, borrowed capital. If the total investment return earned on the leveraged portfolio, R_P, exceeds the borrowing cost on debt, r_D, taking on leverage increases the return on the portfolio. Denoting the return on a leveraged portfolio as R_L, then the return can be calculated as follows:

$$R_L = \frac{Portfolio\ return}{Portfolio\ equity} = \frac{[R_P \times (V_E + V_B) - (V_B \times r_D)]}{V_E} = 15$$

where V_E is the equity of the portfolio and V_B is the debt or borrowed funds. If $R_P < r_D$ then leverage decreases R_L.

For example, for a leveraged EUR10 million equity portfolio that generates an 8 percent total investment return, R_P, over one year and is financed 30 percent with debt at 5 percent, then the leveraged return, R_L, is:

$$R_L = R_p + \frac{V_B}{V_E}(R_p - r_D) = 8\% + \frac{EUR3\ million}{EUR7\ million}(8\% - 5\%) = 8\%$$

EXAMPLE 15

Return Calculations

An analyst observes the following historic asset class geometric returns:

Exhibit 17: Asset Class Geometric Return

Asset Class	Geometric Return (%)
Equities	8.0
Corporate Bonds	6.5
Treasury bills	2.5

1. The real rate of return for equities is *closest* to:

A. 5.4 percent.
B. 5.8 percent.
C. 5.9 percent.

Hide Solution

Solution:

B is correct. The real rate of return for equities is calculated as follows:

(1 + 0.080)/(1 + 0.0210) − 1 = 5.8%.

2. The real rate of return for corporate bonds is *closest* to:

A. 4.3 percent.
B. 4.4 percent.
C. 4.5 percent.

Hide Solution

Solution:

A is correct. The real rate of return for corporate bonds is calculated as follows:

(1 + 0.065)/(1 + 0.0210) − 1 = 4.3%.

3. The risk premium for equities is closest to:

A. 5.4 percent.
B. 5.5 percent.
C. 5.6 percent.

Hide Solution

Solution:

A is correct. The risk premium for equities is calculated as follows:

(1 + 0.080)/(1 + 0.0250) − 1 = 5.4%.

4. The risk premium for corporate bonds is *closest* to:

A. 3.5 percent.
B. 3.9 percent.
C. 4.0 percent.

Hide Solution

Solution:

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions

(1 + 0.0650)/(1 + 0.0250) − 1 = 3.9%.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

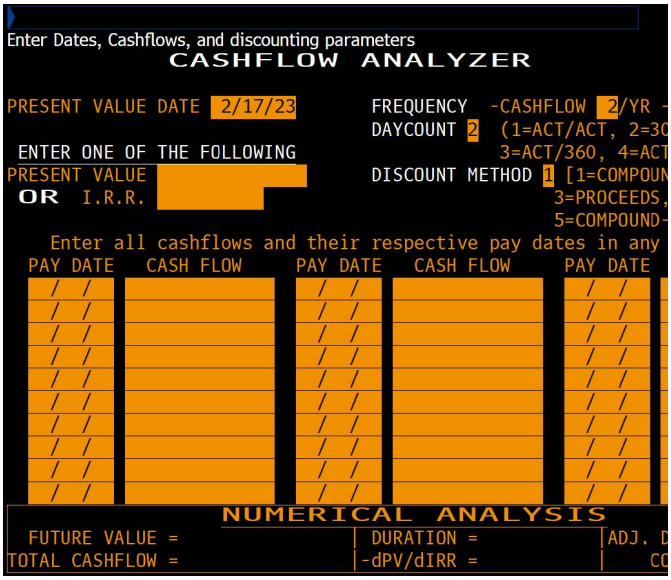
Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{PF} CASHFLOW ANALYZER

The Bloomberg PF function allows you to calculate the internal rate of return on a set of cashflows.

> "PF allows you to customize the methods used for discounting cash flows, day type and compounding frequency so that you can calculate the present value or internal rate of return for an arbitrary set of cash flows and their respective pay dates; helping you to make more informed decisions." -Bloomberg

To explore this function, type PF into the Bloomberg command line and press Enter or <GO>.



© 2023 Bloomberg L.P.

Please provide feedback on this Bloomberg feature.

Discuss Filter

YF

Noticed the adjustment of equation 14 from https://www.cfainstitute.org/sites/default/files/docs/programs/cfa-program/candidate-resources/2025-cfa-li-curriculum-errata-notice.pdf, anyone know why?

Created 4 months ago by Yang Fan — 1 reply | Last Activity: 3 months ago

Show All replies Reply to this Comment

RG

Does anyone know how I can navigate back to Example 8? It would have been more thoughtful if they had included a hyperlink.

Created 4 months ago by Roja Ghasemi — 2 replies | Last Activity: 3 months ago

Show All replies Reply to this Comment

XL

If you don't know which chapter the exam question is from, how can you decide which formula to use? Like example 15, (1+ nominal risk- free rate)=(1 + real risk- free rate)(1 + inflation premium) or nominal risk- free rate=real risk- free rate + inflation premium?

Created 6 months ago by Xintong Li — 2 replies | Last Activity: 4 months ago

Show All replies Reply to this Comment

HW

Expenses including management fees, custody fees, and taxes are NOT included in the GROSS return because they may be different for different investors

Created 5 months ago by Hui Wang — 1 reply | Last Activity: 5 months ago

Show All replies Reply to this Comment

SD

If the real rate of return for equities in example 15 is 5.8, why is the risk premium 5.4%? Shouldn't the equation be (1 + 0.058)/(1 + 0.0250) − 1 instead of (1 + 0.080)/(1 + 0.0250) − 1?

Created 8 months ago by Samuel de Sousa Dias Martins Bettini — 9 replies | Last Activity: 6 months ago

Show All replies Reply to this Comment

MP

Why equity is referred actually as NOTIONAL? Equity is basically what you have and notional what you are exposed to in teh market, in one of the examples exposes as 10 million, equity, financed 30% by debt, therefore what i would understand is you borrow 3 million (getting a notional of 13 mill) but looks like this is not recognized as such and when CFA speaks about equity, it is actually talking about notional. Another twist minder added to the currency pairs quoted the opposite as normal.

Created 6 months ago by Manuel Paredes Caldito — 1 reply | Last Activity: 6 months ago

Show All replies Reply to this Comment

AS

Any Idea how to access bloomberg terminal without paying extra bucks? There should have been some provisions from CFA institute for us ?

Created 6 months ago by Ajay Saxena — 0 replies | Last Activity: 6 months ago

Reply to this Comment

HA

Totally agreed with SD, in the example 15, it should be (1 + 0.058)/(1 + 0.0250) − 1 instead of (1 + 0.080)/(1 + 0.0250) − 1, the practice questions also evidence the same. When requesting to calculate the Risk premium, the numerator is the Real rate of return as well, not the Nominal rate of return.

Created 7 months ago by HUYNH ANH TUAN TRAN — 1 reply | Last Activity: 6 months ago

Show All replies Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.